

Problem Set 2 - IO

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Question 1

Exploring the Data

```
# Load necessary libraries  
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(ggplot2)  
library(readr)  
library(lubridate)
```

Attaching package: 'lubridate'

The following objects are masked from 'package:base':

date, intersect, setdiff, union

```
library(here)
```

here() starts at /Users/tate/SchoolWork

```
# Load the dataset
data <- read_csv(here("Y2S1/I0/ps2/data", "prod_data_all.csv"))
```

Rows: 57748 Columns: 9

```
-- Column specification -----
Delimiter: ","
chr (2): industry, country
dbl (7): id, year, Y, Y2, L, K, M
```

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
# Clean dataset to harmonize year discrepancy - 1981 = 81
data <- data %>%
  mutate(year = ifelse(year < 100, year + 1900, year))

# Display the first few rows of the dataset
head(data)
```

```
# A tibble: 6 x 9
   id year    Y    Y2    L    K    M industry country
  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <chr>    <chr>
1 10250 1981 296. 113. 2.86 15.5 183. metal    col
2 10732 1989 4733. 2566. 86.2 470. 2167. metal    col
3 10732 1990 4800. 2608. 73.6 644. 2192. metal    col
4 10962 1982 7090. 1836. 37.9 5538. 5254. metal    col
5 10962 1983 4942. 1343. 40.5 6314. 3599. metal    col
6 11192 1981 1813. 1311. 24.0 1723. 502. metal    col
```

1.1

```
# Summary Statistics of the data
data %>%
  group_by(country, industry) %>%
  summarize(
    total_obs_sect = n_distinct(id),
    .groups = 'drop'
  ) %>%
  ungroup() %>%
  arrange(country, industry)
```

```
# A tibble: 10 x 3
  country industry total_obs_sect
  <chr>    <chr>          <int>
1 chile   apparel           856
2 chile   food             2584
3 chile   metal            1038
4 chile   text              836
5 chile   wood              867
6 col     apparel           744
7 col     food              908
8 col     metal             632
9 col     text              475
10 col    wood              173
```

```
data %>%
  group_by(year, industry) %>%
  summarize(
    n_obs_t = n_distinct(id),
    .groups = 'drop'
  ) %>%
  ungroup() %>%
  arrange(year, industry) %>%
  print()
```

```
# A tibble: 90 x 3
  year industry n_obs_t
  <dbl> <chr>      <int>
1  1979 apparel     354
2  1979 food      1171
3  1979 metal     373
4  1979 text       391
```

```

5 1979 wood      340
6 1980 apparel   317
7 1980 food     1067
8 1980 metal     350
9 1980 text      356
10 1980 wood     306
# i 80 more rows

```

Chile: 1. Food - 2584, 2. Metal - 1038, 3. Wood - 867, 4. Apparel - 856, 5. Textile - 836

Colombia: 1. Food - 908, 2. Apparel - 744, 3. Metal - 632, 4. Textile - 475, 5. Wood - 173

Apparel Over Time - 1979: 354 → 1996: 292 - Peak at 1981 - 979

Food Over Time - 1979: 1171 → 1996: 1178 - Peak at 1981 - 1951

Metal Over Time - 1979: 373 → 1996: 443 - Peak at 1981 - 890

Textiles Over Time - 1979: 391 → 1996: 304 - Peak at 1981 - 752

Wood Over Time - 1979: 340 → 1996: 290 - Peak at 1981 - 465

1.2

```

# Creation of Investment Variable with 10 per depreciation rate
data <- data %>%
  arrange(id, year) %>%
  group_by(id) %>%
  mutate(
    dep      = 0.10,
    k_next   = lead(K),
    i_raw    = k_next - (1 - dep) * K,
    log_i    = ifelse(i_raw > 0, log(i_raw), NA_real_),
    log_L    = ifelse(L > 0, log(L), NA_real_),
    log_K    = ifelse(K > 0, log(K), NA_real_),
    log_Y2   = ifelse(Y2 > 0, log(Y2), NA_real_)
  ) %>%
  ungroup()

```

Warning: There were 2813 warnings in `mutate()`.

The first warning was:

i In argument: `log\_i = ifelse(i\_raw > 0, log(i\_raw), NA\_real\_)`.

i In group 3: `id = 54`.

Caused by warning in `log()`:

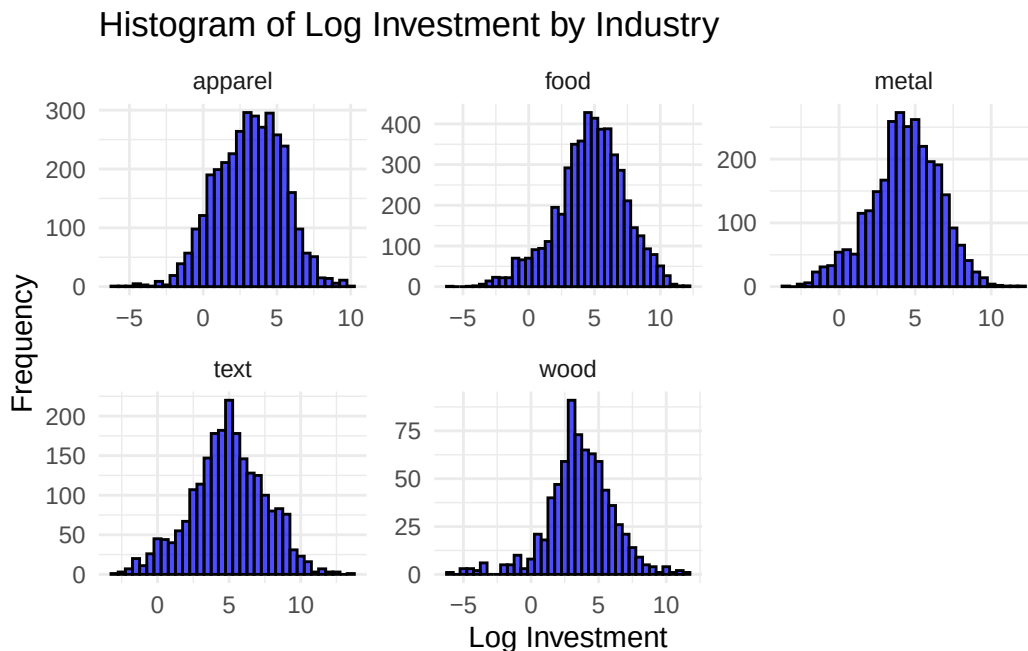
! NaNs produced

i Run `dplyr::last\_dplyr\_warnings()` to see the 2812 remaining warnings.

```
data_col <- data %>% filter(country == "col")
data_chi <- data %>% filter(country == "chile")

# Visualize with Histogram for each industry
ggplot(data_col, aes(x = log_i)) +
  geom_histogram(binwidth = 0.5, fill = "blue", color = "black", alpha = 0.7) +
  facet_wrap(~ industry, scales = "free") +
  labs(title = "Histogram of Log Investment by Industry",
       x = "Log Investment",
       y = "Frequency") +
  theme_minimal()
```

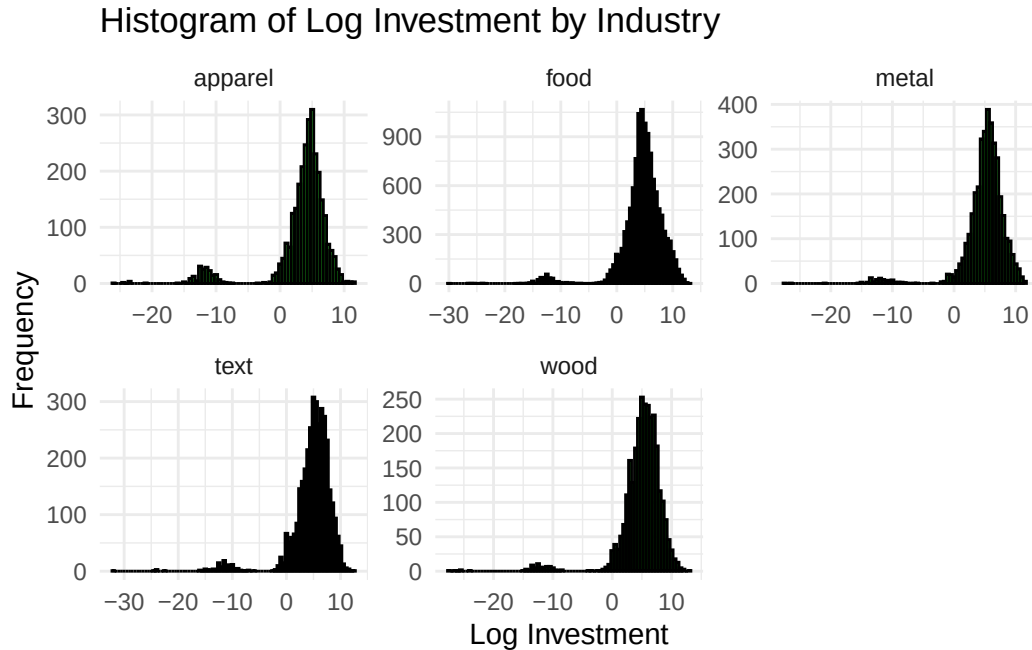
Warning: Removed 3768 rows containing non-finite outside the scale range
(`stat\_bin()`).



```
ggplot(data_chi, aes(x = log_i)) +
  geom_histogram(binwidth = 0.5, fill = "green", color = "black", alpha = 0.7) +
  facet_wrap(~ industry, scales = "free") +
```

```
labs(title = "Histogram of Log Investment by Industry",
     x = "Log Investment",
     y = "Frequency") +
theme_minimal()
```

Warning: Removed 14540 rows containing non-finite outside the scale range (`stat\_bin()`).



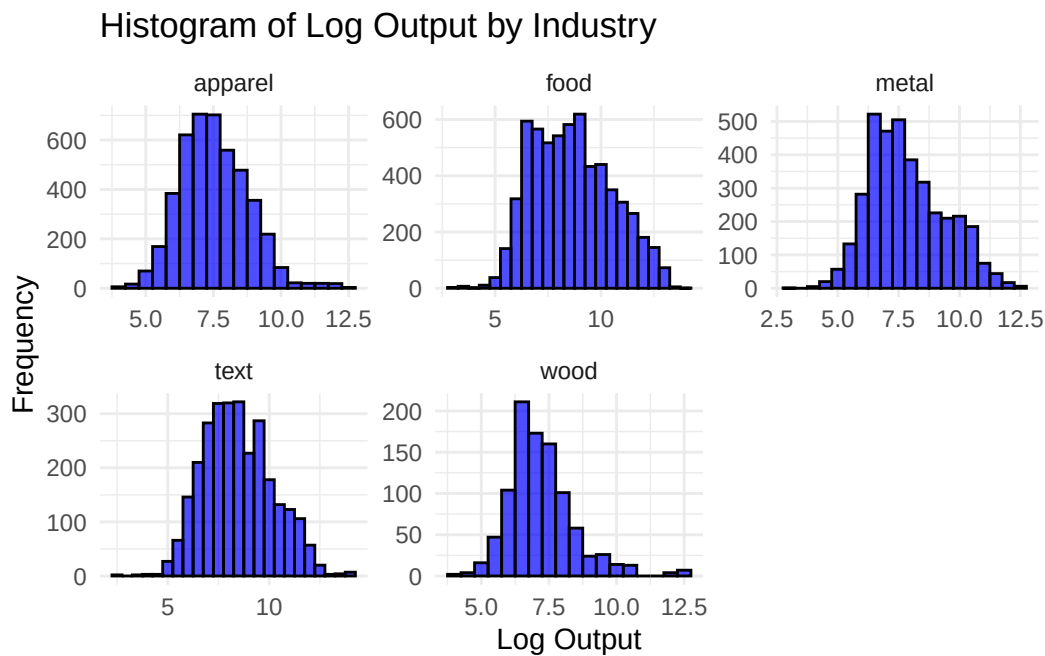
From these graphs, it seems that most industries have similar levels of investment, though textiles, wood, and metal appear to have slightly higher investment levels on average, while apparel is lower.

1.3

```
data_col <- data_col %>%
  group_by(industry) %>%
  mutate(
    log_Y = log(Y),
  ) %>%
  ungroup()
```

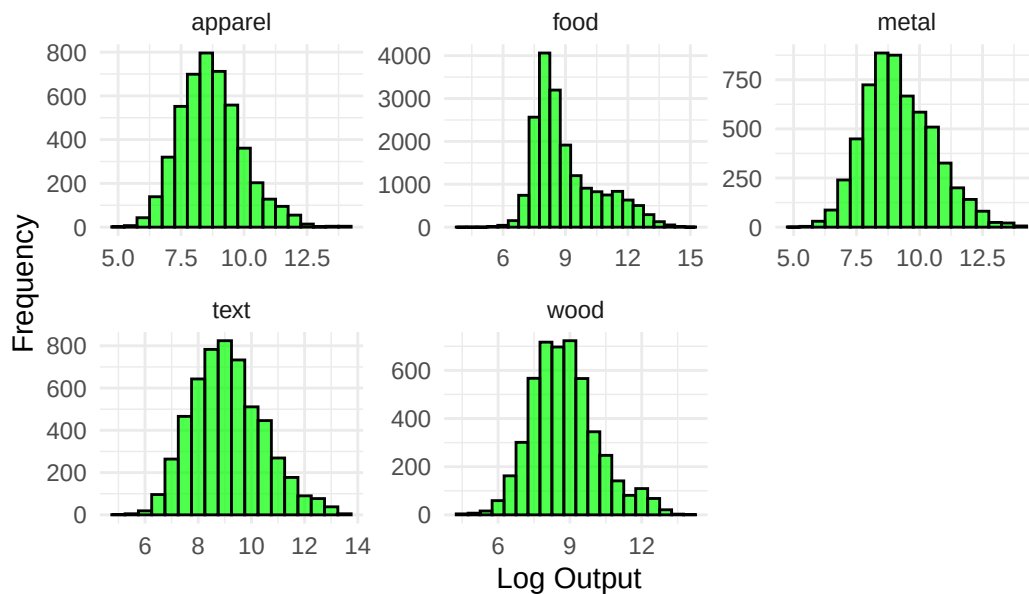
```
data_chi <- data_chi %>%
  group_by(industry) %>%
  mutate(
    log_Y = log(Y),
  ) %>%
  ungroup()

# Visualize with Histogram for each industry
ggplot(data_chi, aes(x = log_Y)) +
  geom_histogram(binwidth = 0.5, fill = "blue", color = "black", alpha = 0.7) +
  facet_wrap(~ industry, scales = "free") +
  labs(title = "Histogram of Log Output by Industry",
       x = "Log Output",
       y = "Frequency") +
  theme_minimal()
```



```
ggplot(data_chi, aes(x = log_Y)) +
  geom_histogram(binwidth = 0.5, fill = "green", color = "black", alpha = 0.7) +
  facet_wrap(~ industry, scales = "free") +
  labs(title = "Histogram of Log Output by Industry",
       x = "Log Output",
       y = "Frequency") +
  theme_minimal()
```

Histogram of Log Output by Industry



1.4

```
# Scatter plot of log_Y and log_L for each sector in each country
data_col <- data_col %>%
  group_by(industry) %>%
  mutate(
    log_L = log(L),
  ) %>%
  ungroup()
data_chi <- data_chi %>%
  group_by(industry) %>%
  mutate(
    log_L = log(L),
  ) %>%
  ungroup()

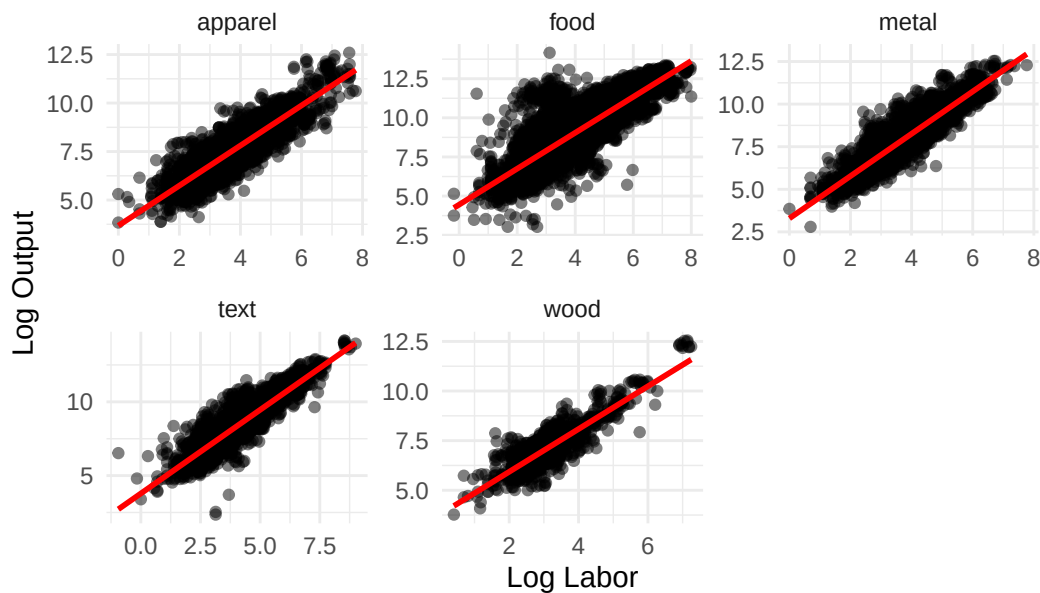
ggplot(data_col, aes(x = log_L, y = log_Y)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", color = "red") +
  facet_wrap(~ industry, scales = "free") +
  labs(title = "Scatter Plot of Log Output vs Log Labor (Colombia)",
       x = "Log Labor",
```



```
y = "Log Output") +  
theme_minimal()
```

`geom\_smooth()` using formula = 'y ~ x'

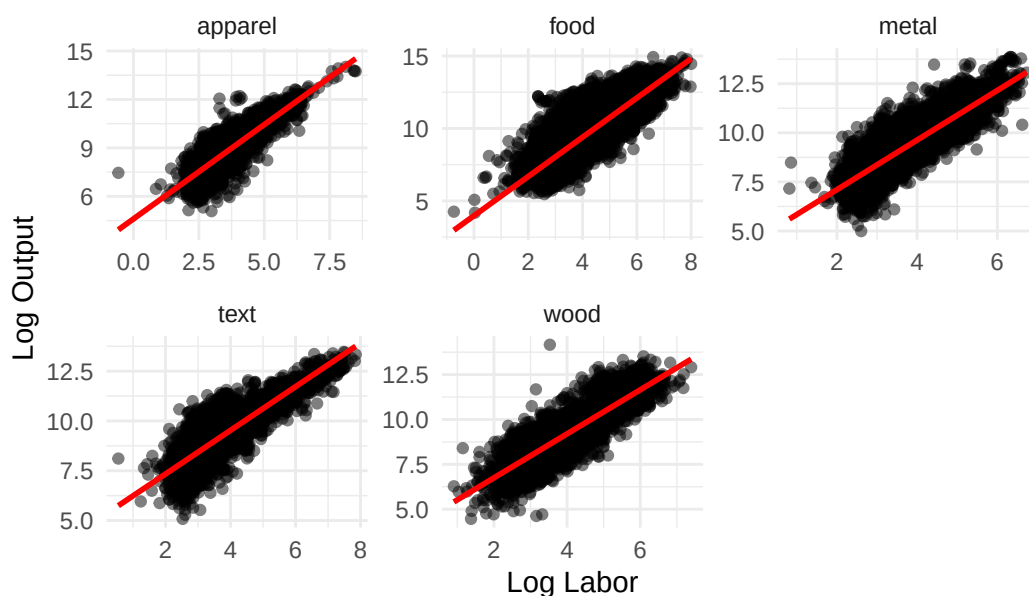
Scatter Plot of Log Output vs Log Labor (Colombia)



```
ggplot(data_chi, aes(x = log_L, y = log_Y)) +  
  geom_point(alpha = 0.5) +  
  geom_smooth(method = "lm", color = "red") +  
  facet_wrap(~ industry, scales = "free") +  
  labs(title = "Scatter Plot of Log Output vs Log Labor (Chile)",  
        x = "Log Labor",  
        y = "Log Output") +  
  theme_minimal()
```

`geom\_smooth()` using formula = 'y ~ x'

Scatter Plot of Log Output vs Log Labor (Chile)



Question 2

2.1

The estimating equation is as follows:

$$\log(Y_{it}) = \alpha_0 + \omega_{it} + \epsilon_{it} + \alpha_L \log(L_{it}) + \alpha_K \log(K_{it})$$

To get here, we take the log of:

$$Y_{it} = e^{\alpha_0 + \omega_{it} + \epsilon_{it}} L_{it}^{\alpha_L} K_{it}^{\alpha_K}$$

Where Y_{it} is the value added output of firm i at time t , L_{it} is labor input and K_{it} is capital input. The ω_{it} term is heterogeneous productivity shocks, varying across firms and years. As discussed in class, we can use OLS to estimate, but we will have some issues with endogeneity via ω_{it} showing up in the labor FOC.

```
# Estimate homogenous parameteres using OLS for each country
ols_col <- lm(log(Y2) ~ log_L + log(K), data = data_col)
summary(ols_col)
```

```
Call:
lm(formula = log(Y2) ~ log_L + log(K), data = data_col)

Residuals:
    Min       1Q   Median       3Q      Max
-5.9761 -0.3644 -0.0486  0.3176  6.1163

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.186356   0.017899  122.15  <2e-16 ***
log_L         0.830586   0.006148  135.09  <2e-16 ***
log(K)        0.272688   0.003876   70.35  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6748 on 18080 degrees of freedom
Multiple R-squared:  0.8233,    Adjusted R-squared:  0.8233
F-statistic: 4.211e+04 on 2 and 18080 DF,  p-value: < 2.2e-16
```

```
ols_chi <- lm(log(Y2) ~ log_L + log(K), data = data_chi)
summary(ols_chi)
```

```
Call:
lm(formula = log(Y2) ~ log_L + log(K), data = data_chi)

Residuals:
    Min       1Q   Median       3Q      Max
-8.6282 -0.5256  0.0572  0.6113  3.9301

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.351275   0.020126  116.83  <2e-16 ***
log_L         0.887047   0.007147  124.12  <2e-16 ***
log(K)        0.302505   0.003518   85.99  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9656 on 39662 degrees of freedom
Multiple R-squared:  0.6596,    Adjusted R-squared:  0.6596
F-statistic: 3.843e+04 on 2 and 39662 DF,  p-value: < 2.2e-16
```

```
# Estimate varying parameters across industries using OLS for each country
form <- as.formula("log(Y2) ~ log_L + log(K) + industry")

ols_ind_col <- lm(form, data = data_col)
summary(ols_ind_col)
```

Call:

```
lm(formula = form, data = data_col)
```

Residuals:

Min	1Q	Median	3Q	Max
-6.0523	-0.3609	-0.0432	0.3252	6.0812

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.106372	0.018968	111.051	< 2e-16 ***
log_L	0.862934	0.006437	134.064	< 2e-16 ***
log(K)	0.244184	0.004173	58.509	< 2e-16 ***
industryfood	0.256587	0.014253	18.002	< 2e-16 ***
industrymetal	0.124933	0.015448	8.087	6.49e-16 ***
industrytext	0.175090	0.016655	10.513	< 2e-16 ***
industrywood	0.058733	0.024148	2.432	0.015 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6685 on 18076 degrees of freedom

Multiple R-squared: 0.8266, Adjusted R-squared: 0.8266

F-statistic: 1.437e+04 on 6 and 18076 DF, p-value: < 2.2e-16

```
ols_ind_chi <- lm(form, data = data_chi)
summary(ols_ind_chi)
```

Call:

```
lm(formula = form, data = data_chi)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.3117	-0.4836	0.0475	0.5572	4.3977

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	2.446987	0.022427	109.110	< 2e-16	***
log_L	0.926433	0.006987	132.587	< 2e-16	***
log(K)	0.280980	0.003468	81.019	< 2e-16	***
industryfood	-0.238625	0.015194	-15.705	< 2e-16	***
industry metal	0.435739	0.018315	23.792	< 2e-16	***
industrytext	0.096545	0.018706	5.161	2.47e-07	***
industrywood	-0.416309	0.019125	-21.768	< 2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9292 on 39658 degrees of freedom

Multiple R-squared: 0.6848, Adjusted R-squared: 0.6848

F-statistic: 1.436e+04 on 6 and 39658 DF, p-value: < 2.2e-16

As can be seen in the regression outputs, when we add heterogeneity across industries, the estimates for labor and capital change slightly. However, we can now ascertain differences in productivity across industries via the industry coefficients.

2.2

```
form_hom <- as.formula("log(Y2) ~ log_L + log(K) + log_i")
form_het <- as.formula("log(Y2) ~ log_L + log(K) + log_i + industry")

ols_inv_col <- lm(form_hom, data = data_col)
summary(ols_inv_col)
```

Call:

```
lm(formula = form_hom, data = data_col)
```

Residuals:

Min	1Q	Median	3Q	Max
-4.8632	-0.3651	-0.0553	0.3107	6.0805

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	2.281928	0.022725	100.416	< 2e-16	***
log_L	0.827831	0.006712	123.337	< 2e-16	***

```
log(K)      0.242230  0.005785  41.871  < 2e-16 ***
log_i      0.030692  0.003856   7.959  1.87e-15 ***
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6355 on 14311 degrees of freedom
 (3768 observations deleted due to missingness)
 Multiple R-squared: 0.8361, Adjusted R-squared: 0.836
 F-statistic: 2.433e+04 on 3 and 14311 DF, p-value: < 2.2e-16

```
ols_inv_chi <- lm(form_hom, data = data_chi)
summary(ols_inv_chi)
```

Call:
 lm(formula = form_hom, data = data_chi)

Residuals:

Min	1Q	Median	3Q	Max
-8.6518	-0.5069	0.0514	0.5866	4.1135

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.520696	0.025843	97.539	<2e-16 ***
log_L	0.844182	0.008600	98.163	<2e-16 ***
log(K)	0.298895	0.004756	62.841	<2e-16 ***
log_i	0.016628	0.001837	9.053	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9253 on 25121 degrees of freedom
 (14540 observations deleted due to missingness)
 Multiple R-squared: 0.6895, Adjusted R-squared: 0.6895
 F-statistic: 1.859e+04 on 3 and 25121 DF, p-value: < 2.2e-16

```
ols_inv_ind_col <- lm(form_het, data = data_col)
summary(ols_inv_ind_col)
```

Call:
 lm(formula = form_het, data = data_col)

Residuals:

Min	1Q	Median	3Q	Max
-4.8049	-0.3610	-0.0461	0.3138	6.0505

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.192015	0.023851	91.903	< 2e-16 ***
log_L	0.861901	0.007056	122.150	< 2e-16 ***
log(K)	0.217762	0.005938	36.675	< 2e-16 ***
log_i	0.026804	0.003828	7.002	2.64e-12 ***
industryfood	0.248039	0.015168	16.353	< 2e-16 ***
industry metal	0.122160	0.016503	7.402	1.41e-13 ***
industrytext	0.144812	0.017662	8.199	2.63e-16 ***
industrywood	0.059857	0.025796	2.320	0.0203 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6294 on 14307 degrees of freedom

(3768 observations deleted due to missingness)

Multiple R-squared: 0.8393, Adjusted R-squared: 0.8392

F-statistic: 1.067e+04 on 7 and 14307 DF, p-value: < 2.2e-16

```
ols_inv_ind_chi <- lm(form_het, data = data_chi)
summary(ols_inv_ind_chi)
```

Call:

```
lm(formula = form_het, data = data_chi)
```

Residuals:

Min	1Q	Median	3Q	Max
-8.3590	-0.4694	0.0346	0.5304	3.8485

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.664046	0.028857	92.320	<2e-16 ***
log_L	0.874458	0.008408	104.005	<2e-16 ***
log(K)	0.279467	0.004673	59.798	<2e-16 ***
log_i	0.018358	0.001779	10.317	<2e-16 ***
industryfood	-0.269709	0.018635	-14.473	<2e-16 ***
industry metal	0.349757	0.022331	15.662	<2e-16 ***

```

industrytext    0.019069    0.022612    0.843    0.399
industrywood   -0.424172    0.023670   -17.920    <2e-16 ***
---

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8934 on 25117 degrees of freedom
 (14540 observations deleted due to missingness)
 Multiple R-squared: 0.7106, Adjusted R-squared: 0.7106
 F-statistic: 8812 on 7 and 25117 DF, p-value: < 2.2e-16

When we add investment as a regressor, we see that the coefficients on labor and capital change slightly again. The investment coefficient is positive and significant across all specifications, indicating that higher investment is associated with higher value added output. Industry level heterogeneity remains mostly significant as well.

Comparing to the results from 2.1, inclusion of investment seems to improve model fit slightly, as seen in the adjusted R-squared values, suggesting that investment is a factor worth considering when modeling firm output.

2.3

```
library(plm)
```

Attaching package: 'plm'

The following objects are masked from 'package:dplyr':

between, lag, lead

```

# Prepare data for panel data estimation
pdata_col <- pdata.frame(data_col, index = c("id", "year"))
pdata_chi <- pdata.frame(data_chi, index = c("id", "year"))
# OP Estimation for each country
op_col <- plm(log(Y2) ~ log_L + log(K), data = pdata_col, model = "pooling",
  effect = "individual", method = "ols")
summary(op_col)

```


Pooling Model

Call:

```
plm(formula = log(Y2) ~ log_L + log(K), data = pdata_col, effect = "individual",  
     model = "pooling", method = "ols")
```

Unbalanced Panel: n = 2851, T = 1-11, N = 18083

Residuals:

Min.	1st Qu.	Median	3rd Qu.	Max.
-5.976116	-0.364381	-0.048644	0.317612	6.116349

Coefficients:

	Estimate	Std. Error	t-value	Pr(> t)
(Intercept)	2.1863562	0.0178995	122.146	< 2.2e-16 ***
log_L	0.8305863	0.0061482	135.095	< 2.2e-16 ***
log(K)	0.2726880	0.0038760	70.354	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares: 46592

Residual Sum of Squares: 8234

R-Squared: 0.82328

Adj. R-Squared: 0.82326

F-statistic: 42113.3 on 2 and 18080 DF, p-value: < 2.22e-16

```
op_chi <- plm(log(Y2) ~ log_L + log(K), data = pdata_chi, model = "pooling",  
              effect = "individual", method = "ols")  
summary(op_chi)
```

Pooling Model

Call:

```
plm(formula = log(Y2) ~ log_L + log(K), data = pdata_chi, effect = "individual",  
     model = "pooling", method = "ols")
```

Unbalanced Panel: n = 6012, T = 1-18, N = 39665

Residuals:

Min.	1st Qu.	Median	3rd Qu.	Max.
-8.628234	-0.525594	0.057228	0.611264	3.930117

Coefficients:

	Estimate	Std. Error	t-value	Pr(> t)
(Intercept)	2.3512747	0.0201261	116.827	< 2.2e-16 ***
log_L	0.8870471	0.0071468	124.119	< 2.2e-16 ***
log(K)	0.3025049	0.0035178	85.993	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Total Sum of Squares: 108630

Residual Sum of Squares: 36977

R-Squared: 0.65959

Adj. R-Squared: 0.65958

F-statistic: 38426.1 on 2 and 39662 DF, p-value: < 2.22e-16